

RETAIL SALES ANALYSIS AND INSIGHTS

1. Project Overview and Objective

This project involves cleaning and analyzing retail sales data using Excel and creating an interactive Power BI dashboard. The objective is to analyze retail sales to identify high-performing products, profitable regions, and customer purchasing patterns, enabling data-driven business decisions.

2. Data Sources

- **Source:** Flipkart Dataset
 - **Timeline:** 2015 – 2020
 - **Domain:** Retail / Commerce / Sales Analytics
 - **Format & Size:** Excel / CSV (~100,000 orders)
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3. Problem Statement

- Analyze sales performance across regions and product categories to identify top-performing areas and profit trends.
- Understand customer purchasing behavior to enhance marketing and retention strategies.
- Evaluate delivery performance and order returns for operational improvements.
- Identify profitable products and segments to support strategic business decisions.

Key Business Questions:

- Which products generate the highest revenue and profit?
 - Which regions or delivery zones are most profitable?
 - Which customer segments order most frequently and have the highest ratings?
 - How do delivery types and days affect customer satisfaction and returns?
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4. Attribute (Column / Features) Details

Attribute Name	Type	Description
Order Date	Date	Date when the order was placed
OrderID	Number	Unique ID for each order
Delivery Date	Date	Date when the order was delivered
CustomerID	Number	Unique ID for each customer
Age	Number	Customer age
Gender	Text	Customer gender (M/F)
Location	Text	City or region of customer
Zone	Text	Delivery zone
Delivery Type	Text	Standard / Express / Shipped Abroad
Product Category	Text	Main category of product
Sub Category	Text	Subcategory of product
Unit Price	Number	Price of 1 unit
Shipping Fee	Number	Delivery charges
Quantity	Number	Number of units ordered
Sale Price	Number	Total price paid
Status	Text	Delivered / Returned
Reason	Text	Return reason (if any)
Rating	Number	Customer rating
Total Revenue	Number	Revenue per order
Cost	Number	Cost per order
Profit	Number	Revenue – Cost
Profit%	Number	Profit percentage
AgeGroup	Text	Youth / Adult / Senior
Margin	Text	High / Medium / Low

Calculated Columns:

- **Profit** = **TotalRevenue** – **Cost**
 - **Profit %** = **DIVIDE**([Total Profit], [Total Revenue], 0)
 - **Delivery Days** = **DeliveryDate** – **OrderDate**
 - **AgeGroup**: Youth / Adult / Senior
 - **Margin**: High / Medium / Low
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5. Tools & Technologies

- **Excel**: Data cleaning, transformation, calculated columns, Pivot Tables, trend analysis.
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

1. **Data Cleaning**: Handled missing values, corrected dates, standardized categories.
 2. **Descriptive Analysis**: Calculated Total Revenue, Profit, Profit%, Average Rating, and Delivery Days to understand business performance.
 3. **Segmentation & Insights**: Analyzed data by Product Category, Sub-Category, Zone, Customer Age Group, and Delivery Type to identify top-performing segments.
 4. **Trend Analysis**: Monitored monthly and weekly trends in orders, revenue, and profits to detect patterns and peak periods.
 5. **Pivot Tables**: Created for quick insights into revenue drivers, profit contribution, and customer preferences.
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Key Insights

- Health & Beauty Products and Fashion are top revenue-generating product categories.
 - Zone 3 contributes the highest profit among all regions.
 - Standard delivery Type show higher customer satisfaction compared to standard shipping.
 - Adult customer segment accounts for the highest purchase volume and repeat orders.
 - Peak sales occur during Weekdays
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7. Data Modelling and DAX (Power BI)

Data Model:

- Established relationships between Sales, Product, Delivery, and Customer tables.
- Defined One-to-Many relationships using primary keys.
- Created Lookup Tables for Product Category and Delivery Type.
- Ensured proper cardinality and filter direction.

DAX Measures Implemented:

Total Revenue = SUM(Sales [Total Revenue])

Total Profit = SUM(Sales [Profit])

Total Cost = SUM(Sales[Cost])

Total Orders = DISTINCTCOUNT(Data [OrderID])

Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)

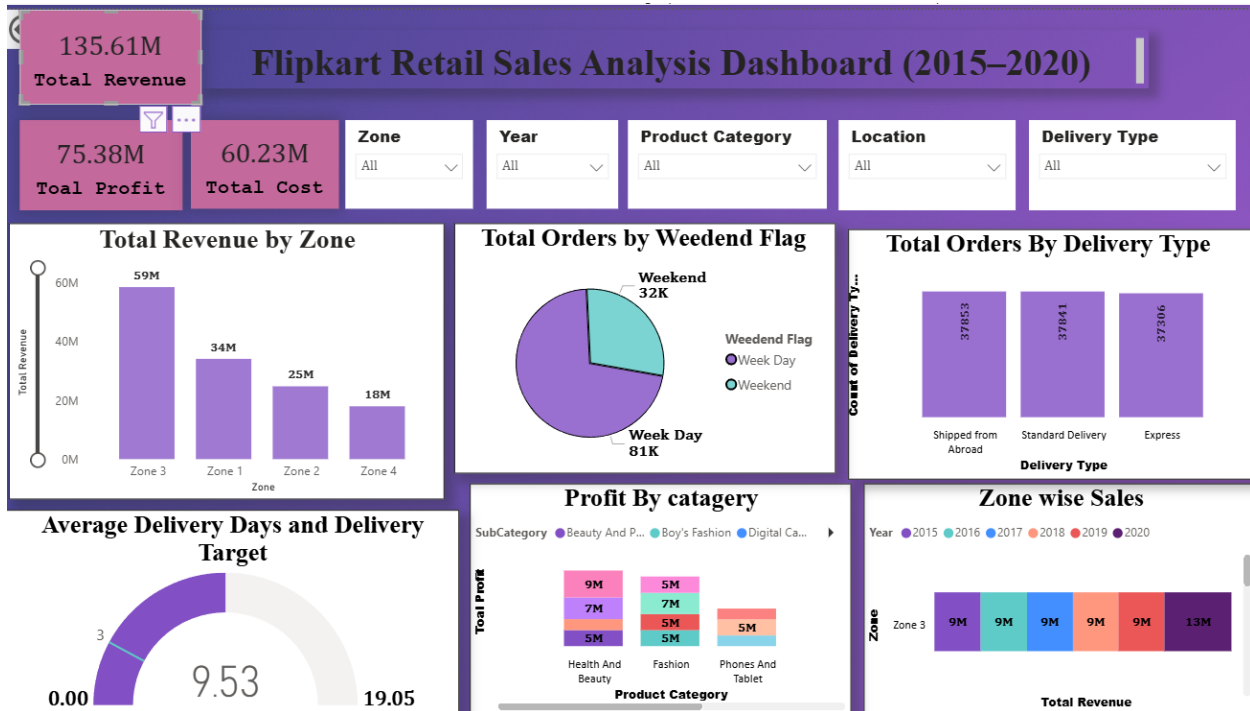
Average Rating = AVERAGE(Data[Rating])

Average Delivery Days = AVERAGE(Data [Delivery Days])

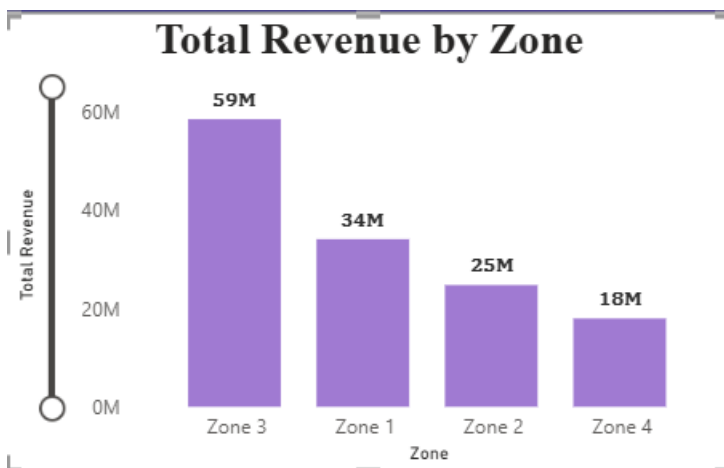
8. Dashboard Features:

Visualizations Used:

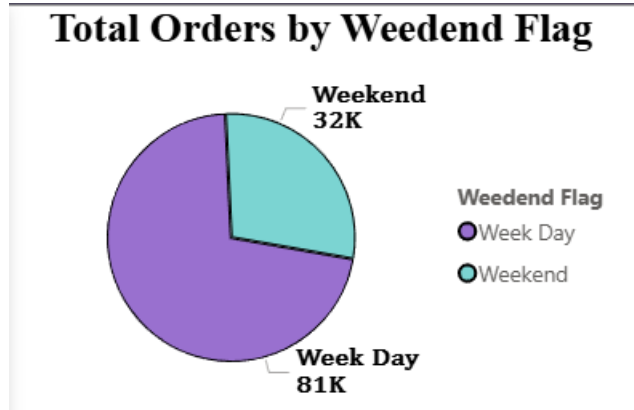
- KPI Cards → Revenue, Profit, Cost



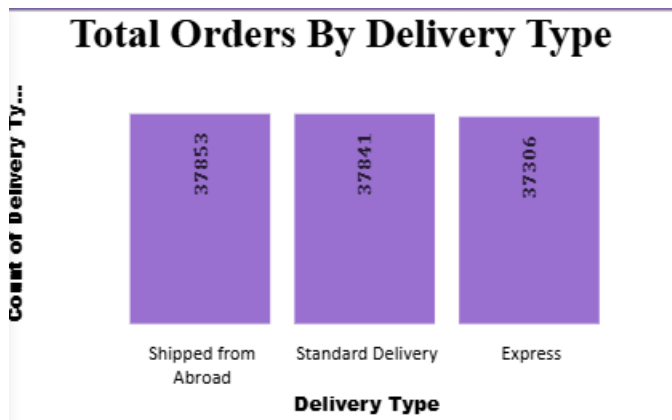
- Clustered Column Chart → Revenue by Zone



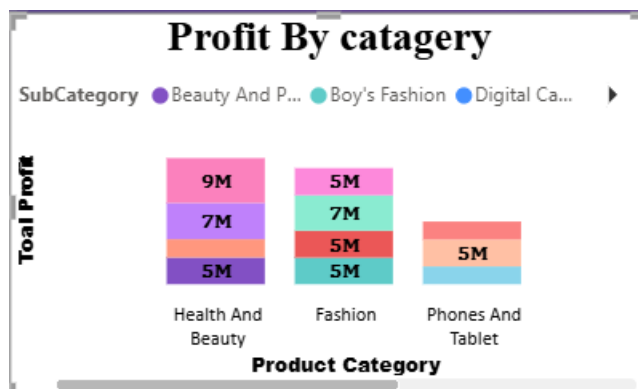
- Pie Chart → Orders by Weekend Flag



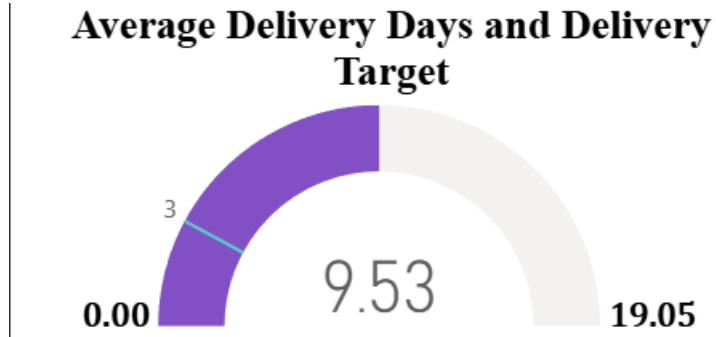
- Column Chart → Orders by Delivery Type



- Stacked Bar Chart → Profit by Category & Subcategory



- Gauge Chart → Average Delivery Days vs Target



- Trend Chart → Zone-wise Yearly Sales



Interactive Features:

- Slicers for Zone, Year, Category, Delivery Type
- Drill-down enabled for Category → Subcategory
- Dynamic filtering across visuals
- Clean titles & data labels
- Unified purple theme for branding

9. Insights & Conclusions

Descriptive Analysis

- Total Revenue: 135.61M
 - Total Profit: 75.38M
 - Zone 3 generated highest revenue (59M).
 - Weekday orders (81K) significantly higher than Weekend (32K).
 - Standard delivery type has highest order volume.
 - Health & Beauty and Fashion are top-performing categories.
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Diagnostic Analysis

- Zone 3 likely has higher customer concentration.
 - Weekday sales dominance indicates business/working-day purchase behaviour.
 - Standard delivery preferred due to cost-effectiveness.
 - Adult segment contributes most due to purchasing power.
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Predictive Analysis

- Revenue expected to grow in Zone 3 if current trend continues.
 - Fashion & Beauty categories will remain dominant.
 - Weekday-focused marketing campaigns will perform better.
 - Express delivery demand may increase in metro zones.
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Prescriptive Analysis

- Increase inventory for high-performing categories.
- Focus marketing budget on Zone 3.
- Offer weekend discounts to improve weekend sales.
- Improve delivery time for Express to increase satisfaction.
- Target adult customers with loyalty programs.

10. Conclusion

The integration of Excel and Power BI enabled a complete end-to-end data analytics workflow—from data cleaning and transformation to advanced visualization and business insight generation.

The dashboard effectively highlights revenue drivers, profitable zones, customer purchasing behavior, and delivery performance. The insights derived support strategic decision-making in marketing, inventory planning, and operational efficiency.

Overall, this project demonstrates strong capabilities in:

- Data Cleaning & Pre-processing
- Data Modelling
- DAX Calculations
- Interactive Dashboard Design
- Business Insight Generation